WORKED SOLUTION

# STEP III 2004, Question 5

Pure • Trigonometric Equations

**STEP III 2004, Question 5**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q5

## The key idea

Two cosines are equal exactly when their angles agree up to sign and a multiple of $2\pi$. Passing to tangents throws away the information that distinguishes $\theta$ from $\theta + \pi$, so the implication only goes one way, and that is the point of the counterexample. In parts (i) and (ii) we put $a\cos x + b\sin x$ into the form $R\cos(x - \alpha)$, use the first result to get $\tan x$ in terms of $\tan\omega$, and then use the sizes of the angles to choose the correct quadrant.

## Solution

### The opening result

If $\cos(x - \alpha) = \cos\beta$, then $x - \alpha = \pm\beta + 2n\pi$ for some integer $n$, since cosine is even and has period $2\pi$. So

$$x = \alpha + \beta + 2n\pi \quad \text{or} \quad x = \alpha - \beta + 2n\pi.$$

Since tan has period $\pi$, and so certainly period $2\pi$, this gives $\tan x = \tan(\alpha + \beta)$ or $\tan x = \tan(\alpha - \beta)$ (whenever these tangents are defined).

The converse fails because tan has period $\pi$ but cos does not. Take $\alpha = \beta = 0$ and $x = \pi$. Then $\tan x = 0 = \tan(\alpha + \beta)$, but

$$\cos(x - \alpha) = \cos\pi = -1 \neq 1 = \cos\beta.$$

In general, $x = \alpha + \beta + \pi$ gives $\cos(x - \alpha) = -\cos\beta$.

Two facts about $\omega$ will be useful. Since $\tan\omega = \frac{4}{3}$, we have $\cos\omega = \frac{3}{5}$ and $\sin\omega = \frac{4}{5}$. Since $\frac{4}{3} > 1$, we also have $\frac{\pi}{4} < \omega < \frac{\pi}{2}$.

**Part (i)**

Here $\sqrt{1^2+7^2} = 5\sqrt{2}$, so we divide by $5\sqrt{2}$:

$$\tfrac{1}{5\sqrt{2}}\cos x - \tfrac{7}{5\sqrt{2}}\sin x = \tfrac{1}{\sqrt{2}}.$$

Let $\varphi$ be the acute angle with $\cos\varphi = \frac{1}{5\sqrt{2}}$ and $\sin\varphi = \frac{7}{5\sqrt{2}}$, so that $\tan\varphi = 7$. The equation becomes

$$\cos(x+\varphi) = \cos\tfrac{\pi}{4},$$

which is the opening result with $\alpha = -\varphi$ and $\beta = \frac{\pi}{4}$. Using $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$,

$$\tan(\alpha+\beta) = \tan(\tfrac{\pi}{4} - \varphi) = \frac{1-7}{1+7} = -\frac{3}{4},$$
$$\tan(\alpha-\beta) = -\tan(\varphi + \tfrac{\pi}{4}) = -\frac{7+1}{1-7} = \frac{4}{3}.$$

**Ask yourself** Each of these tangent values belongs to two angles in $[0, 2\pi]$, but the equation has only two solutions there. Which two are they?

As the counterexample showed, the tangent alone can't decide this, so we go back to $x = \alpha \pm \beta + 2n\pi$ and work out where $\alpha \pm \beta$ actually lies. Since $\tan\varphi = 7 > 1$, we have $\frac{\pi}{4} < \varphi < \frac{\pi}{2}$.

- $\alpha + \beta = \frac{\pi}{4} - \varphi$ lies in $(-\frac{\pi}{4}, 0)$ and has tangent $-\frac{3}{4}$. Now $\tan(\frac{\pi}{2} - \omega) = \cot\omega = \frac{3}{4}$, and $\frac{\pi}{2} - \omega \in (0, \frac{\pi}{4})$. Hence $\alpha + \beta = \omega - \frac{\pi}{2}$, and adding $2\pi$ gives $x = \frac{3\pi}{2} + \omega$.
- $\alpha - \beta = -\varphi - \frac{\pi}{4}$ lies in $(-\frac{3\pi}{4}, -\frac{\pi}{2})$ and has tangent $\frac{4}{3}$. The angle $\omega - \pi$ also lies in this interval and has tangent $\frac{4}{3}$. Tangent is one-to-one on an interval of length less than $\pi$ that avoids its asymptotes, so $\alpha - \beta = \omega - \pi$. Adding $2\pi$ gives $x = \pi + \omega$.

Both values lie in $[0, 2\pi]$ because $0 < \omega < \frac{\pi}{2}$, and no other choice of $n$ gives a value in this range. Therefore

$$x = \pi + \omega \quad \text{or} \quad x = \tfrac{3\pi}{2} + \omega.$$

To confirm, at $x = \pi + \omega$ we get $\cos x - 7\sin x = -\frac{3}{5} + \frac{28}{5} = 5$, and at $x = \frac{3\pi}{2} + \omega$ we have $\cos x = \sin\omega$ and $\sin x = -\cos\omega$, which give $\frac{4}{5} + \frac{21}{5} = 5$.

**Part (ii)**

Here $\sqrt{2^2+11^2} = \sqrt{125} = 5\sqrt{5}$, so

$$\tfrac{2}{5\sqrt{5}}\cos x + \tfrac{11}{5\sqrt{5}}\sin x = \tfrac{2}{\sqrt{5}}.$$

Let $\alpha$ be acute with $\cos\alpha = \frac{2}{5\sqrt{5}}$ and $\sin\alpha = \frac{11}{5\sqrt{5}}$, so $\tan\alpha = \frac{11}{2}$. Let $\beta$ be acute with $\cos\beta = \frac{2}{\sqrt{5}}$, so $\sin\beta = \frac{1}{\sqrt{5}}$ and $\tan\beta = \frac{1}{2}$. The equation is $\cos(x-\alpha) = \cos\beta$, and

$$\tan(\alpha-\beta) = \frac{\frac{11}{2} - \frac{1}{2}}{1 + \frac{11}{4}} = \frac{5}{15/4} = \frac{4}{3},$$
$$\tan(\alpha+\beta) = \frac{\frac{11}{2} + \frac{1}{2}}{1 - \frac{11}{4}} = \frac{6}{-7/4} = -\frac{24}{7}.$$

The second value is $\tan 2\omega$, because

$$\tan 2\omega = \frac{2\cdot\frac{4}{3}}{1 - \frac{16}{9}} = \frac{8/3}{-7/9} = -\frac{24}{7}.$$

Again we place the angles. Both $\alpha$ and $\beta$ are acute, so

- $\alpha - \beta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with tangent $\frac{4}{3}$, so $\alpha - \beta = \omega$;
- $\alpha + \beta \in (0, \pi)$ with negative tangent, so it lies in $(\frac{\pi}{2}, \pi)$. Since $\frac{\pi}{4} < \omega < \frac{\pi}{2}$, the angle $2\omega$ also lies in $(\frac{\pi}{2}, \pi)$ and has the same tangent, so $\alpha + \beta = 2\omega$.

Both $\omega$ and $2\omega$ are in $[0, 2\pi]$, and adding $\pm 2\pi$ takes them outside it. Hence the solutions are

$$x = \omega \quad \text{and} \quad x = 2\omega,$$

and one solution is twice the other, as required. As a direct check, $x = \omega$ gives $\frac{6}{5} + \frac{44}{5} = 10$. For $x = 2\omega$ we have $\cos 2\omega = \frac{9-16}{25} = -\frac{7}{25}$ and $\sin 2\omega = \frac{24}{25}$, which give $-\frac{14}{25} + \frac{264}{25} = 10$.

## Checks

- Each equation should have exactly two roots in $[0, 2\pi]$, because $5 < 5\sqrt{2}$ and $10 < 5\sqrt{5}$, so the right-hand side lies strictly between $-R$ and $R$.
- Numerically, $\omega \approx 0.9273$. A sign-change search finds roots 4.0689 and 5.6397 for (i), which are $\pi + \omega$ and $\frac{3\pi}{2} + \omega$. For (ii) it finds 0.9273 and 1.8546, which are $\omega$ and $2\omega$.

## Common slips

- Solving $\tan x = \frac{4}{3}$ or $\tan x = -\frac{3}{4}$ and listing all four angles in $[0, 2\pi]$. Two of them are spurious, for exactly the reason given by the counterexample.
- Taking $\varphi$ in (i) with the wrong sign, for example writing $\cos(x - \varphi)$ instead of $\cos(x + \varphi)$ for $\cos x - 7\sin x$.

## Beyond STEP

The difficulty here is that tan is not injective, so "$\tan x = \tan y$" only tells you $x$ modulo $\pi$. In complex analysis the same issue appears in the definition of the argument and of branches of $\log z$, where angles are known only up to multiples of $2\pi$. In that setting, $a\cos x + b\sin x = \mathrm{Re}\left((a - \mathrm{i}b)\mathrm{e}^{\mathrm{i}x}\right)$, and $R$ and $\alpha$ are just the modulus and argument of $a + \mathrm{i}b$. Needham's *Visual Complex Analysis* develops this geometric view beautifully.

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